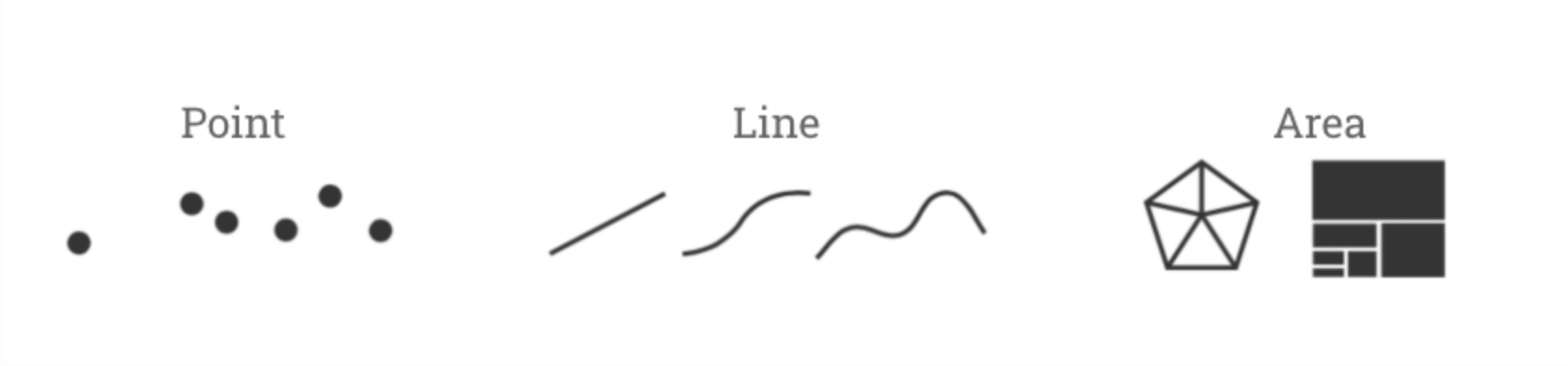
Language of Data Visualization

Marks and Channels



Marks ・　マーク

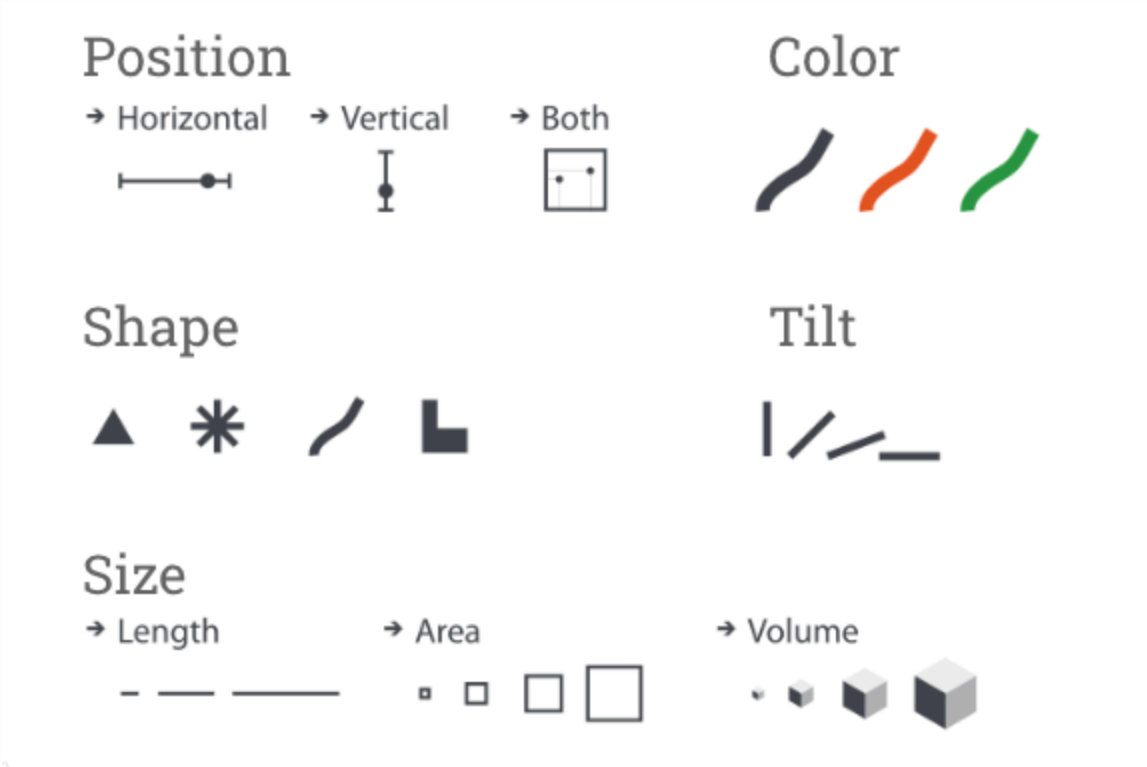
The shapes that represent data.
データを表す形

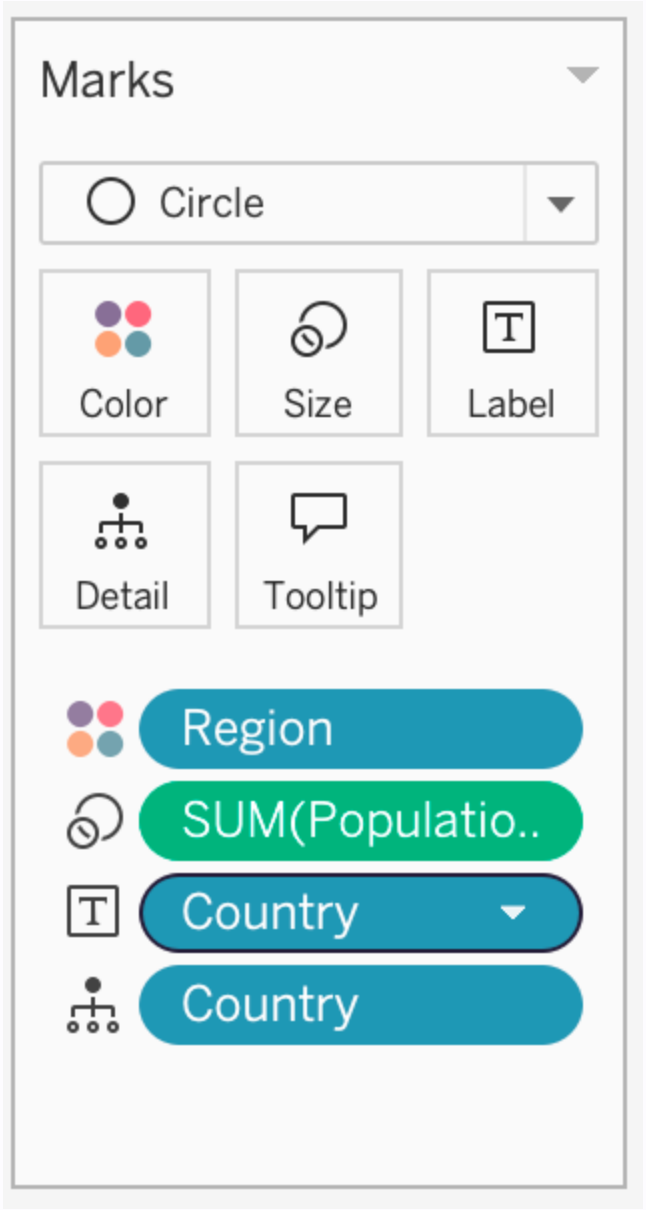


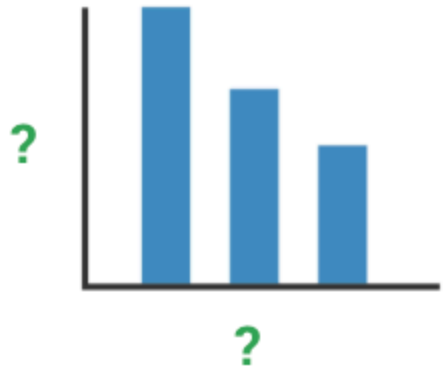
Graphics from Munzner. Visualization Analysis and Design (2015)

Channels ・ チャネル

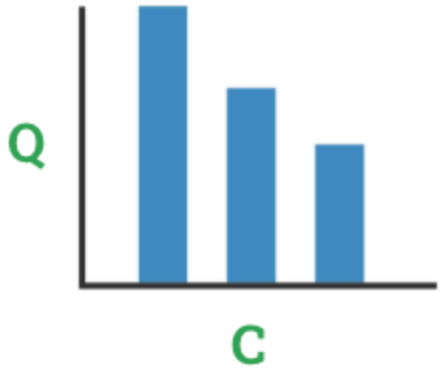
Ways to vary the appearance of marks
マークの見た目を変える方法



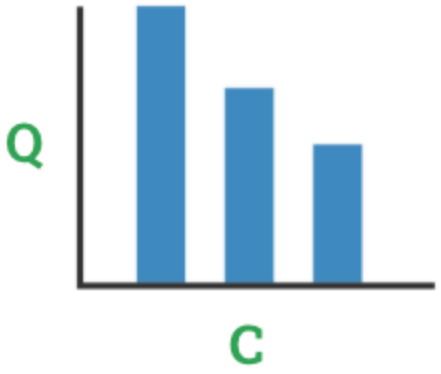

Marks
Circle
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Size
Label
Detail
Tooltip
Region
SUM(Populatio..
Country
Country



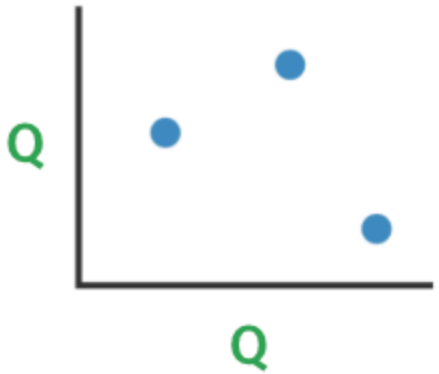
Mark: ?



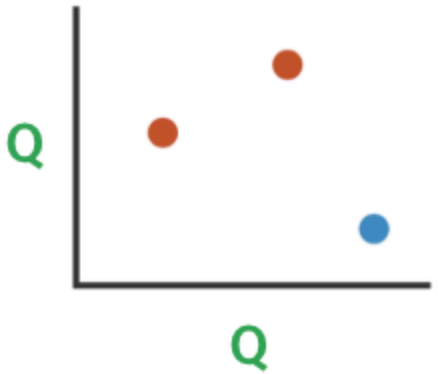
Mark: **line**



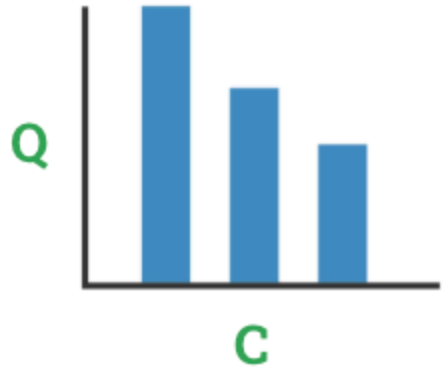
Mark: **line**



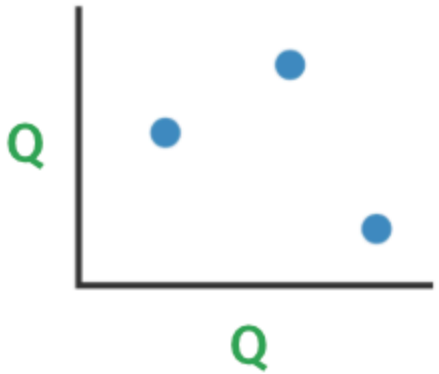
Mark: **point**



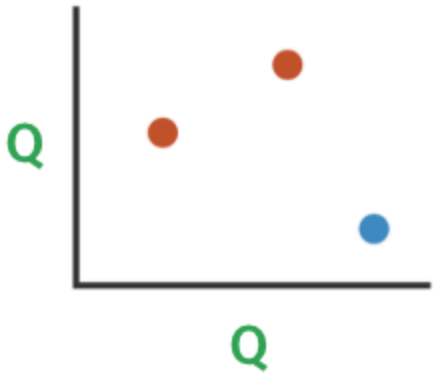
Mark: **point**
Channels:
Color: **C**



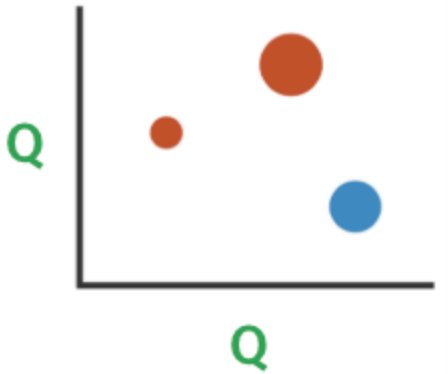
Mark: **line**



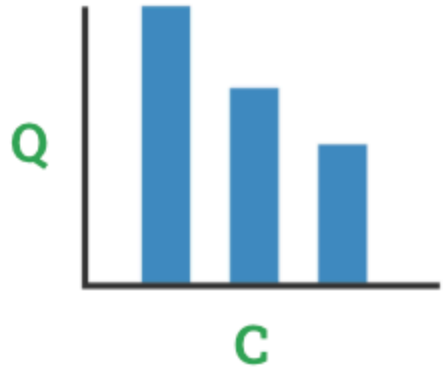
Mark: **point**



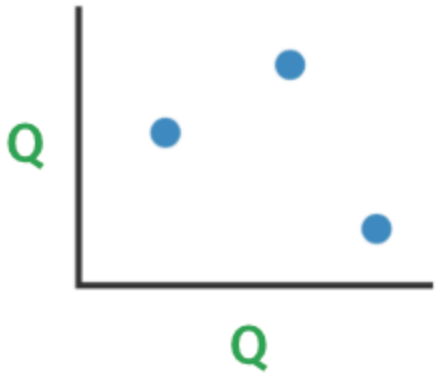
Mark: **point**
Channels:
Color: **C**



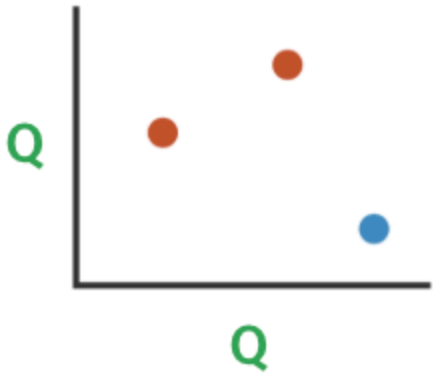
Mark: **point**
Channels: **?**



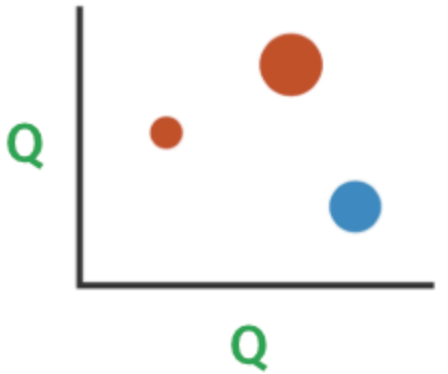
Mark: **line**



Mark: **point**



Mark: **point**
Channels:
Color: **C**



Mark: **point**
Channels:
Color: **C**
Size: **Q**

Position

→ Horizontal → Vertical → Both



Color



Shape



Tilt



Size

→ Length → Area → Volume



Position

→ Horizontal → Vertical → Both



Color



Categorical



Shape



Tilt



×

Ordinal



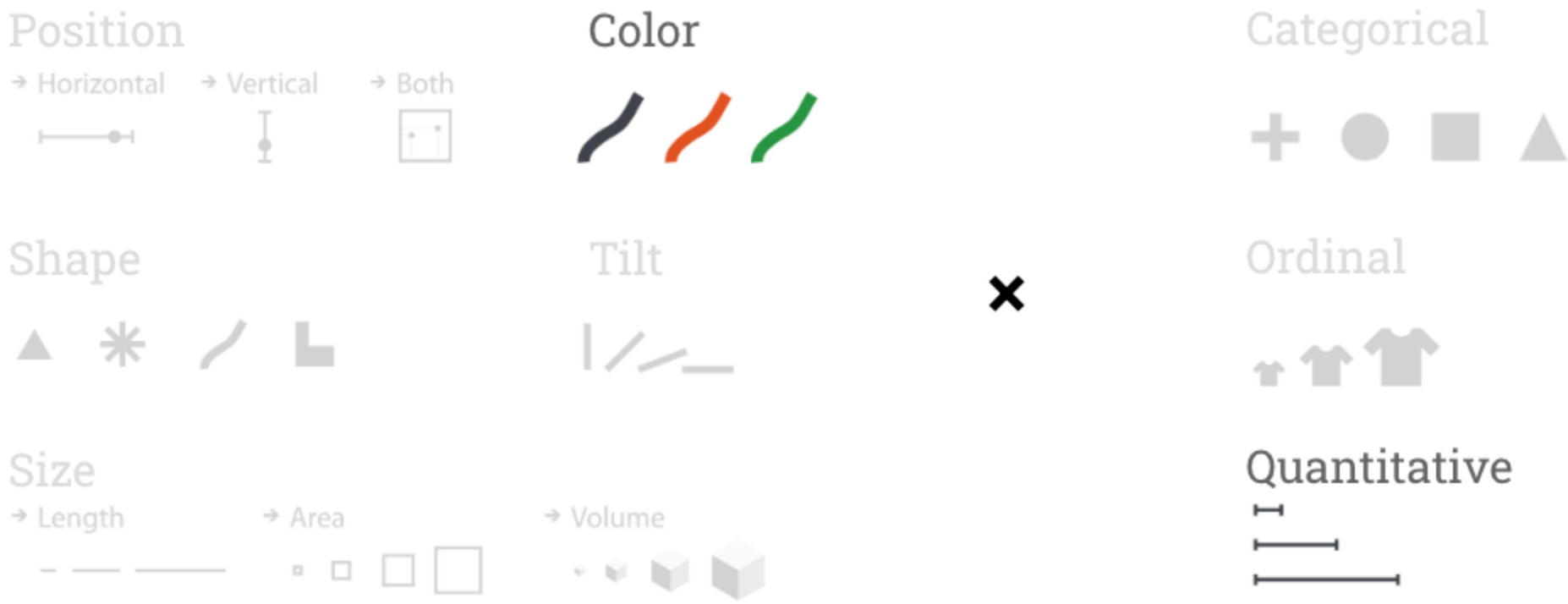
Size

→ Length → Area → Volume



Quantitative




Position
Horizontal
Vertical
Both
Color
Categorical
Shape
Tilt
Ordinal
Size
Length
Area
Volume
Quantitative

Filters

Marks

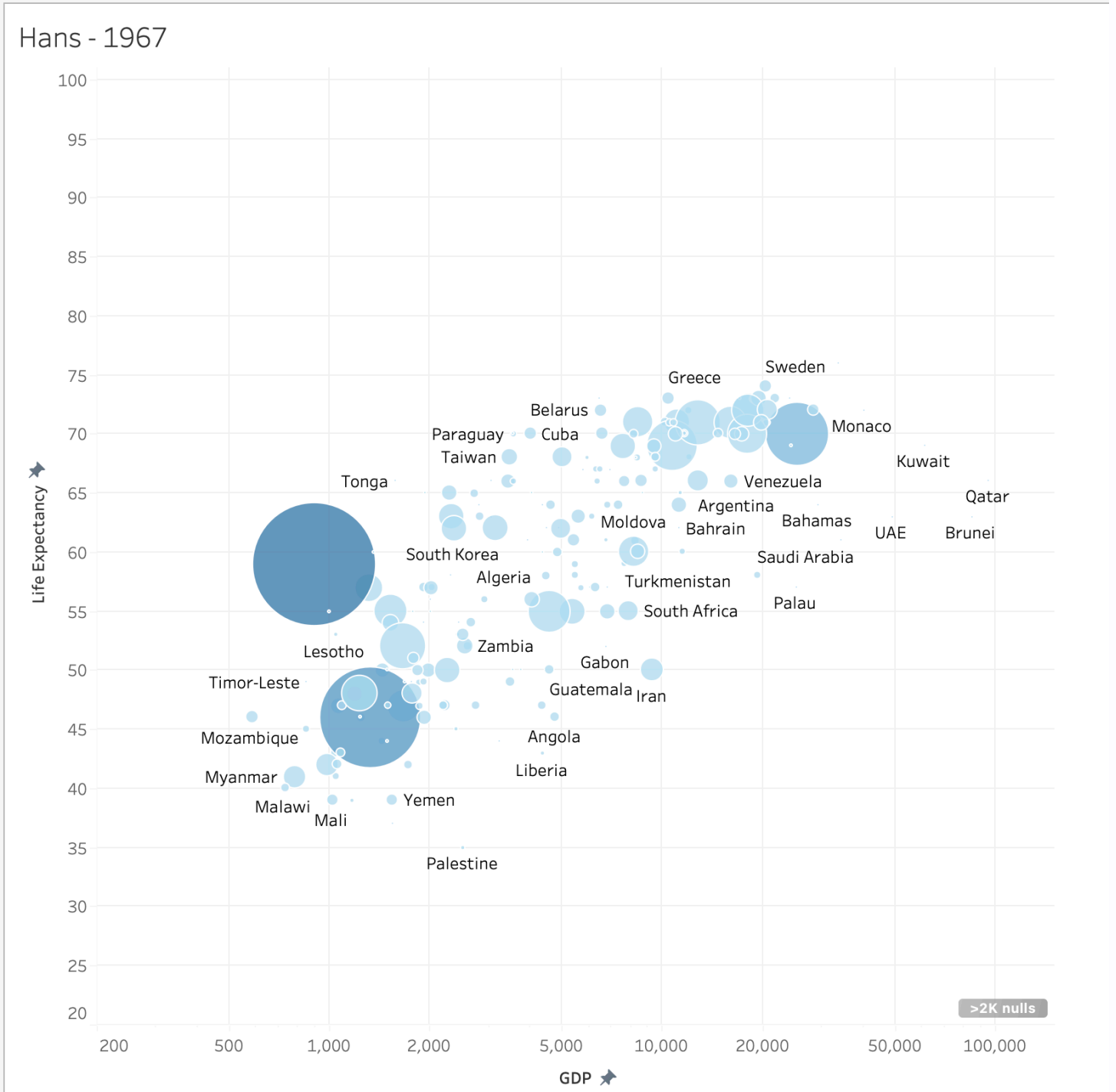
Circle

Color | Size | Label | Detail | Tooltip

SUM(Populatio..
SUM(Populatio..
Country
Country

SUM(Population)

651 999M



Position

→ Horizontal → Vertical → Both

Color



Categorical



Shape

Tilt

×

Ordinal

Size

→ Length → Area → Volume

Quantitative

Filters

Marks

Circle

Color | Size | Label

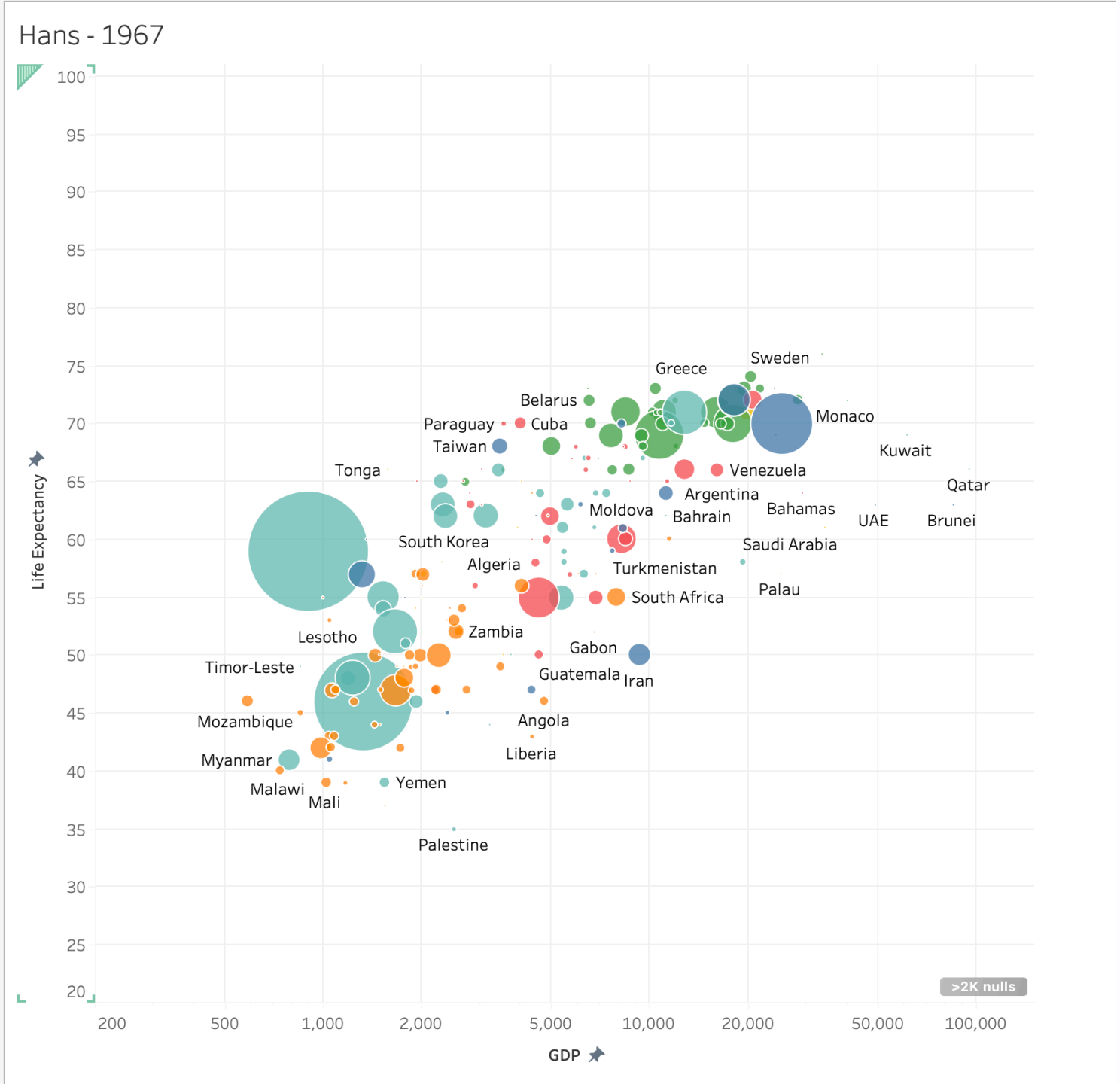
Detail | Tooltip

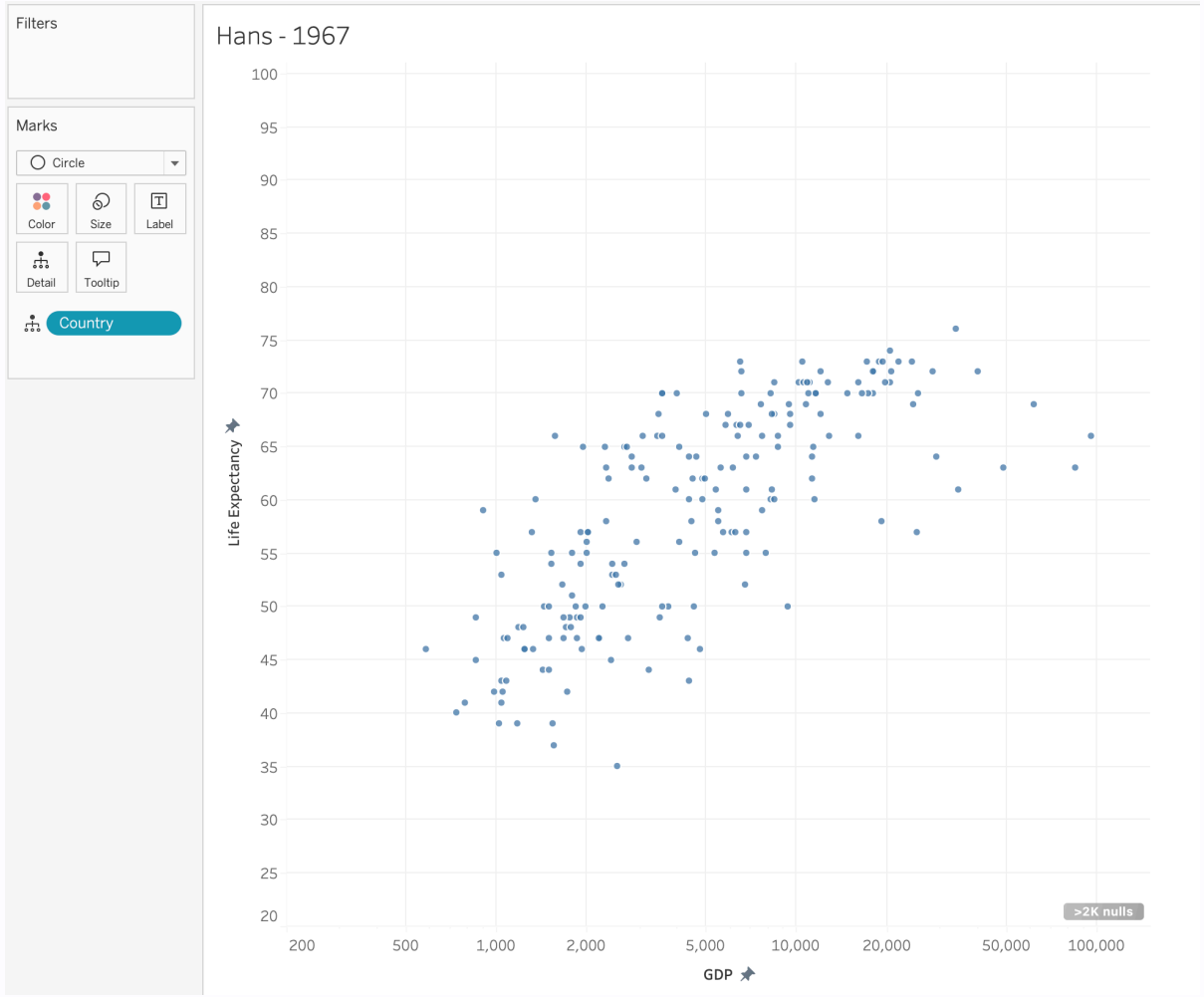
Region

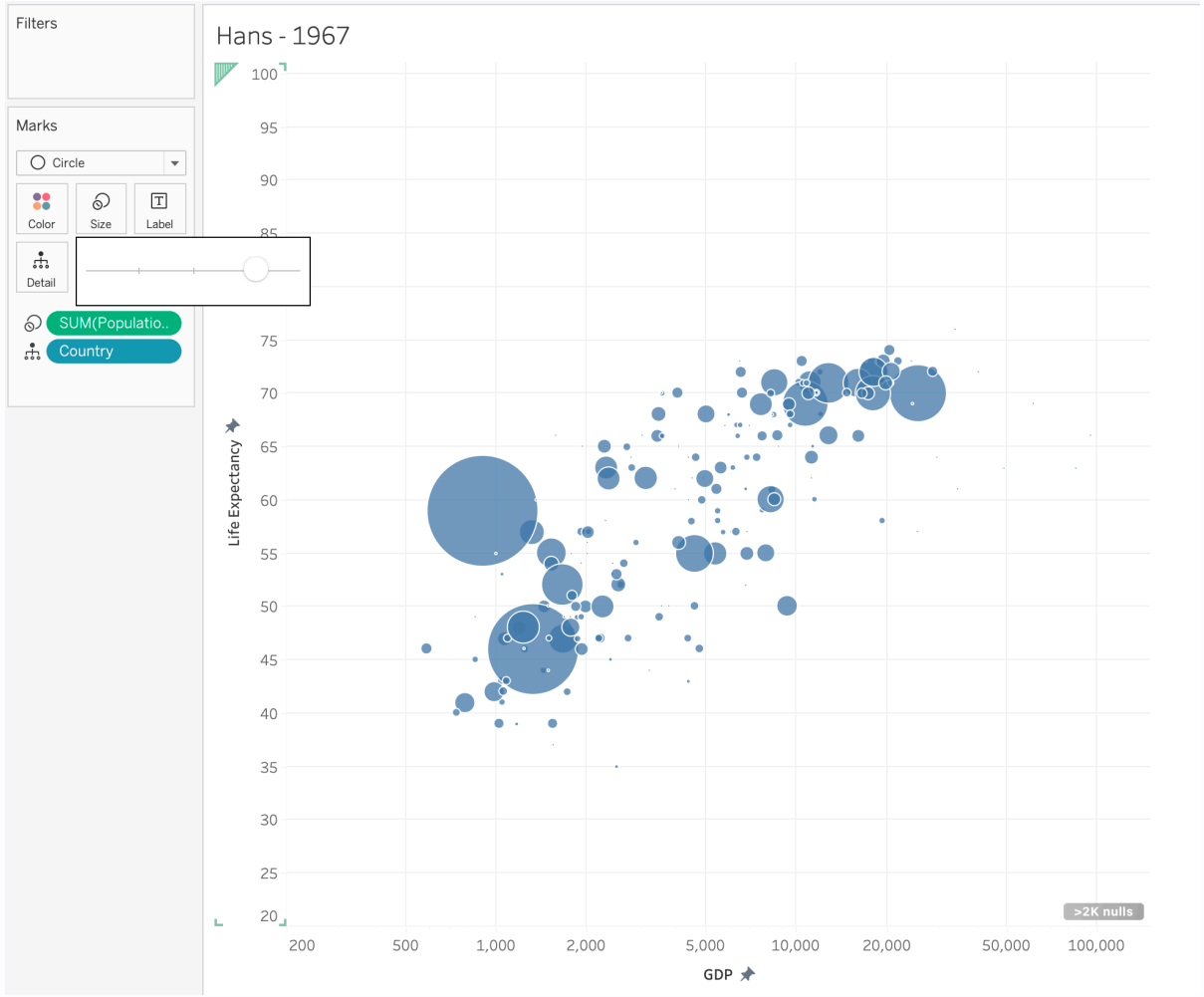
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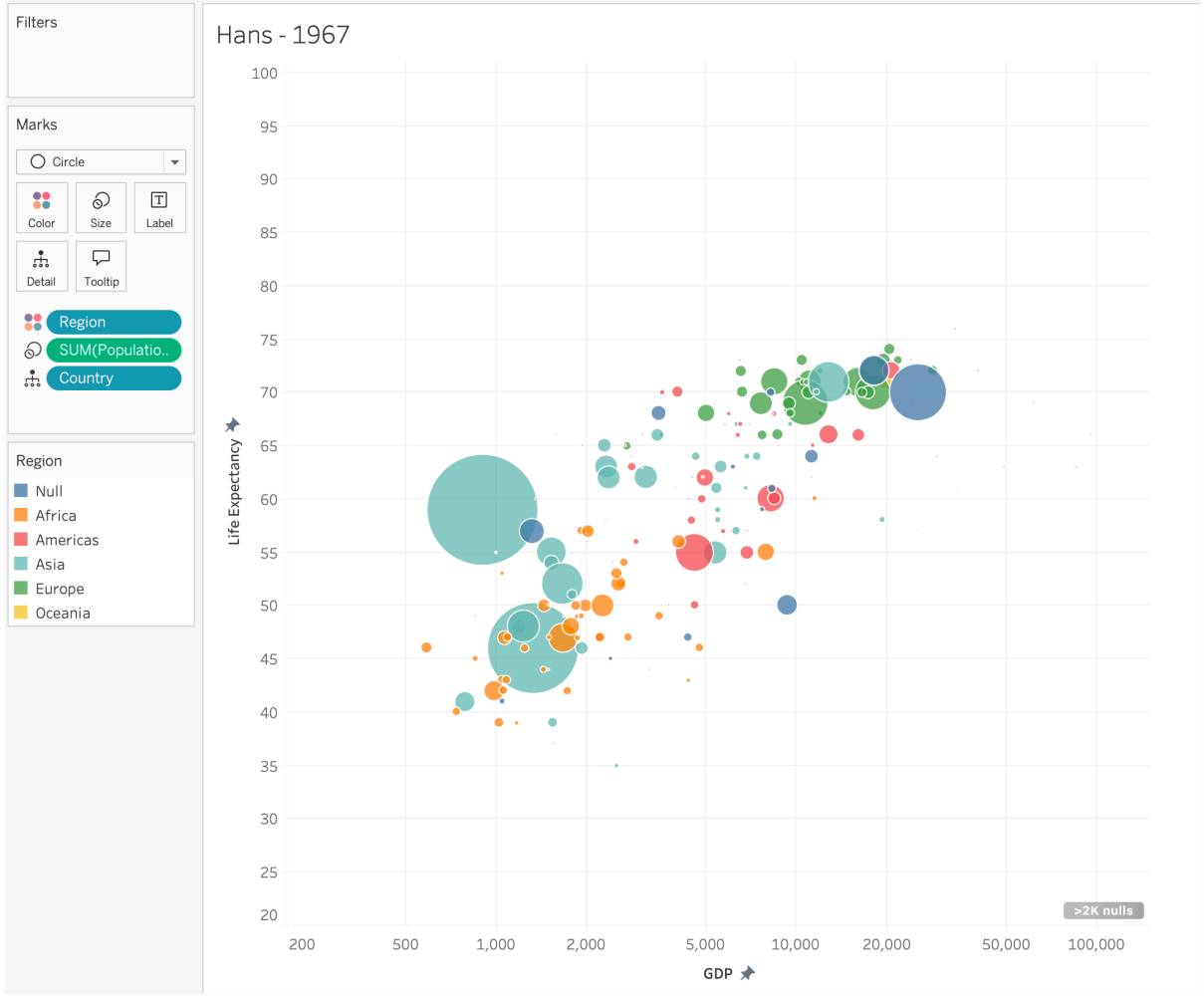
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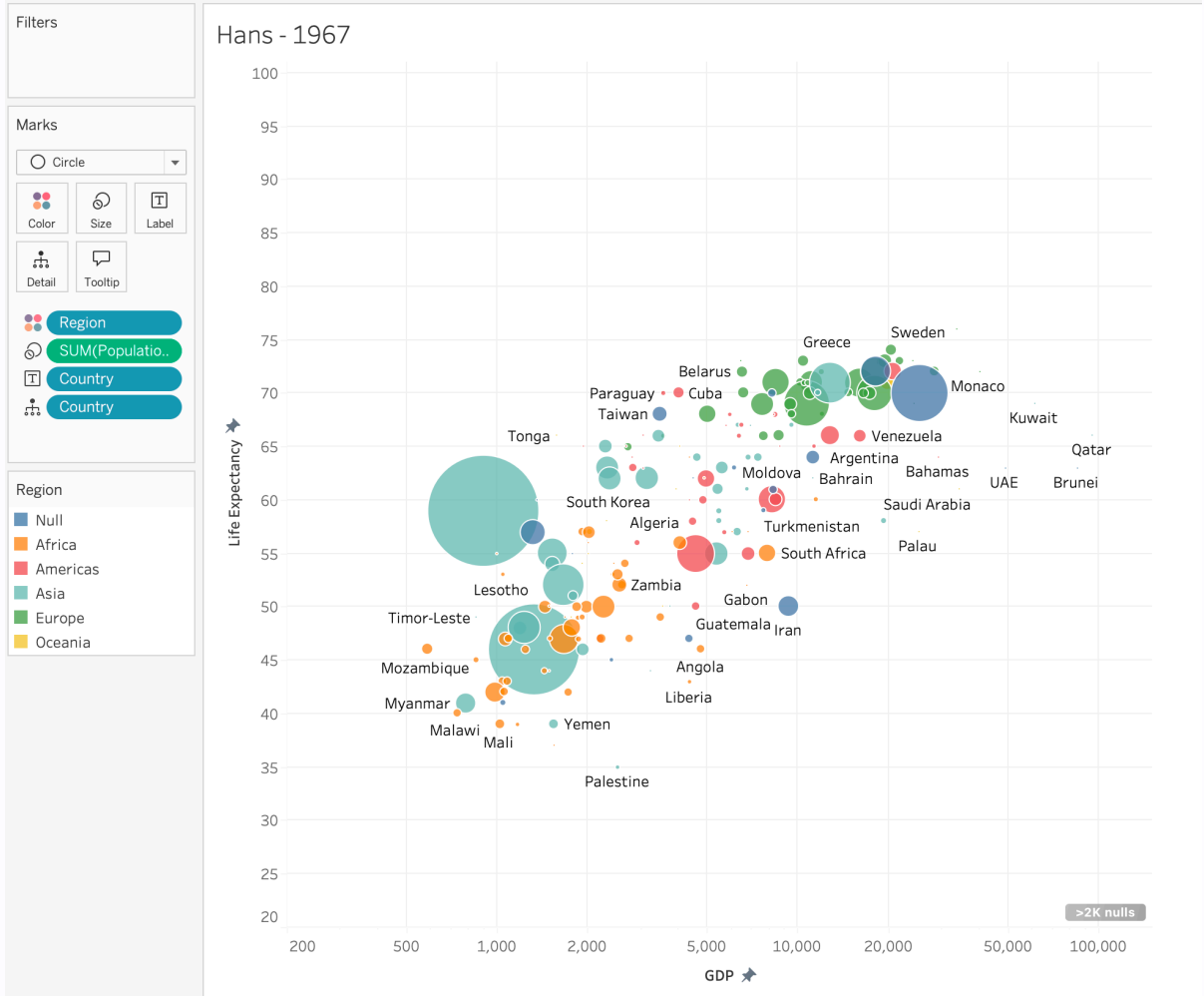
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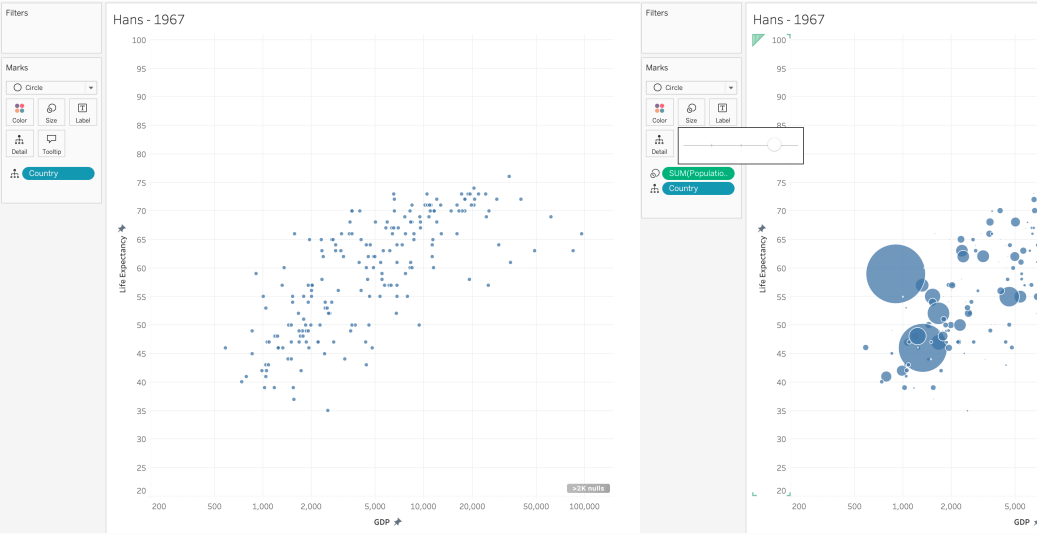


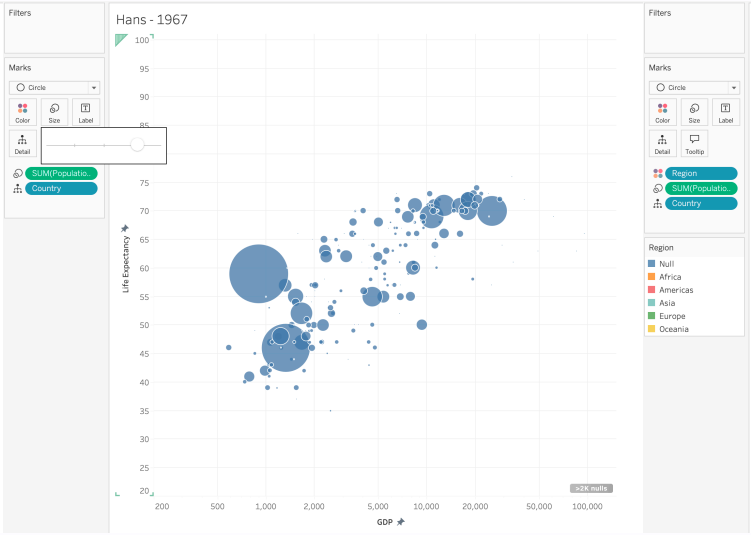

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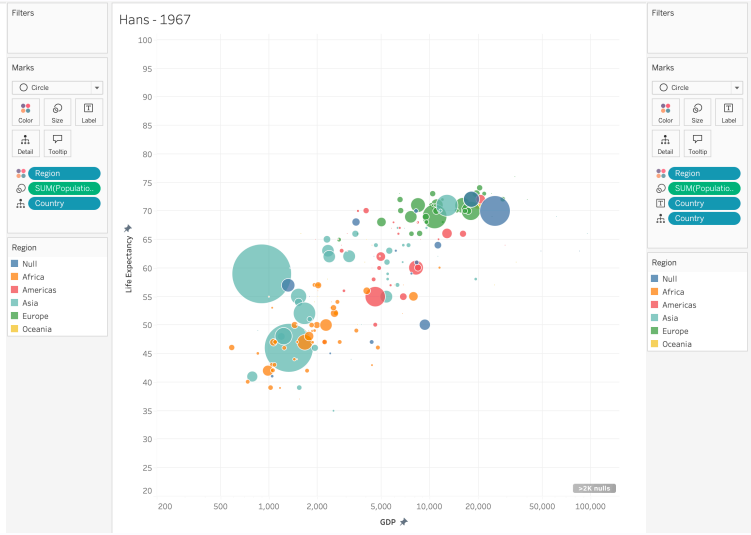

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Life Expectancy
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GDP
>2K nulls

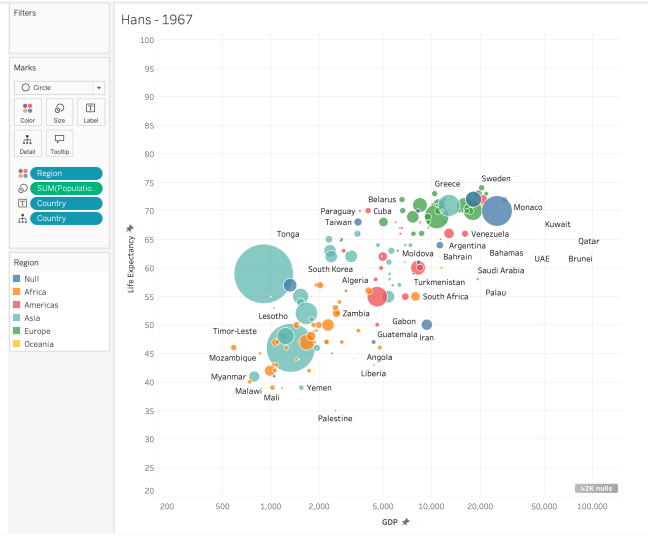

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Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Qatar
Argentina
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
Palau
South Africa
Zambia
Lesotho
Gabon
Timor-Leste
Guatemala
Iran
Mozambique
Angola
Liberia
Myanmar
Malawi
Yemen
Mali
Palestine
>2K nulls


Filters
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Hans - 1967
Life Expectancy
GDP
>2K nulls


Filters
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Hans - 1967
Life Expectancy
GDP
>2K nulls


Filters
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Oceania
Hans - 1967
Life Expectancy
GDP
>2K nulls

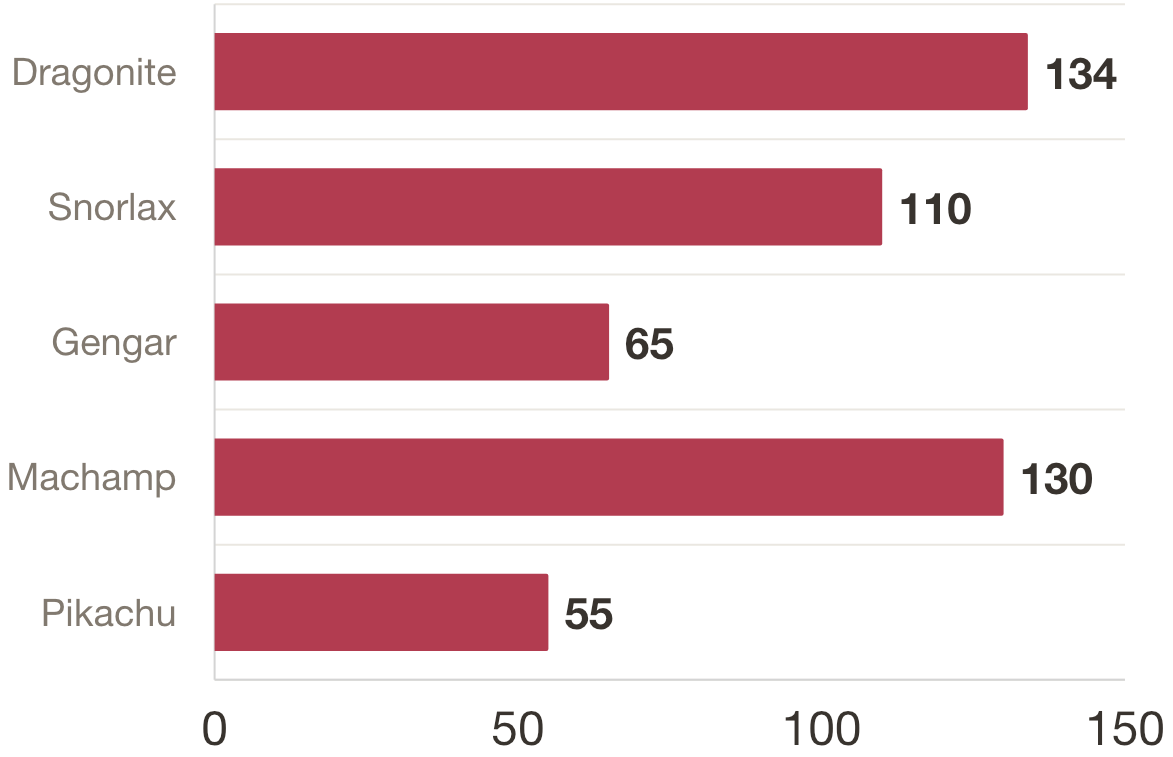
Filters
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Hans - 1967
Life Expectancy
GDP
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Hans - 1967
Life Expectancy
GDP
>2K nulls
Greece
Sweden
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Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Argentina
Qatar
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
Palau
South Africa
Lesotho
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Yemen
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Palestine

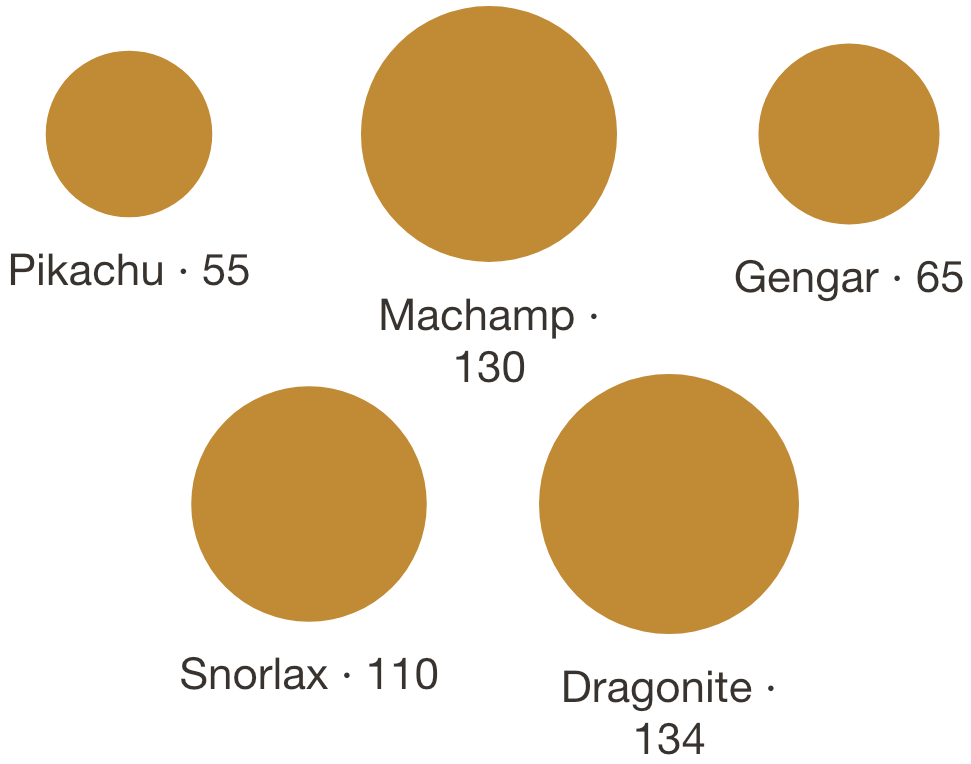
Which comparison is easier?

Same Pokémon. Same Attack values.

POSITION + LENGTH



AREA

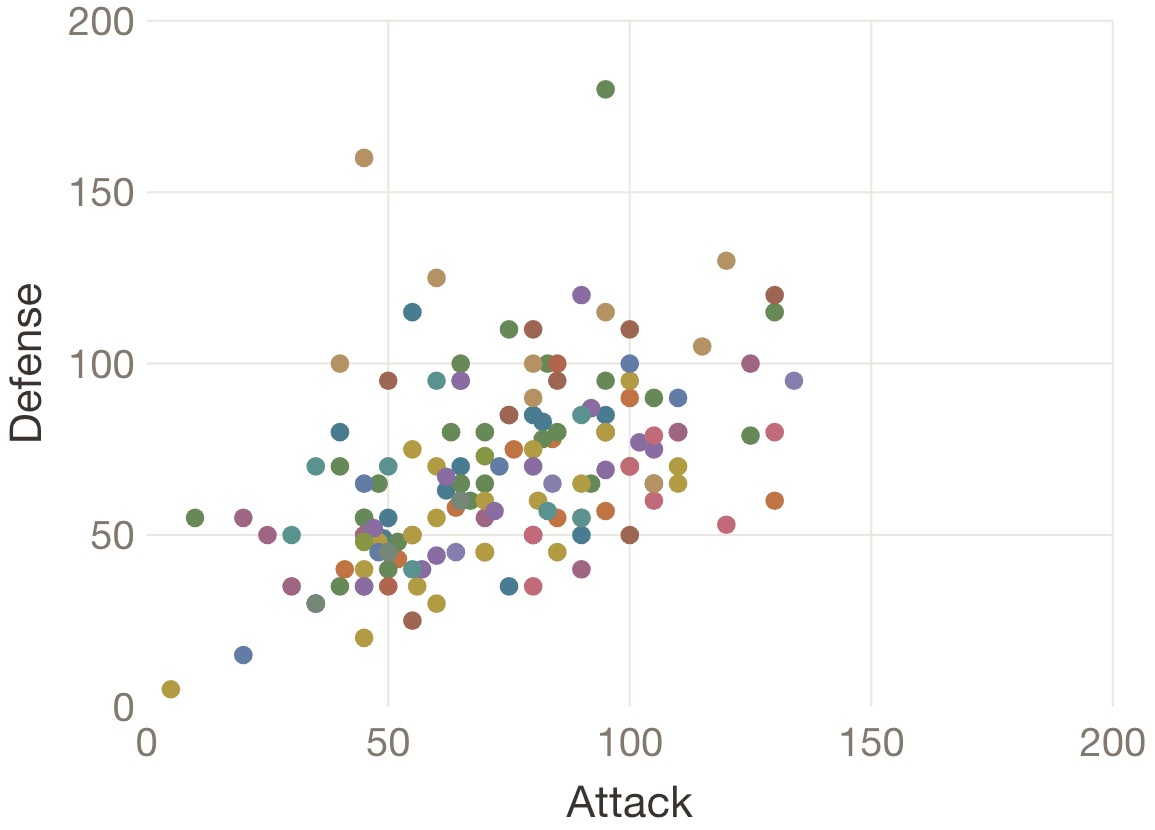


5 selected species · base Attack

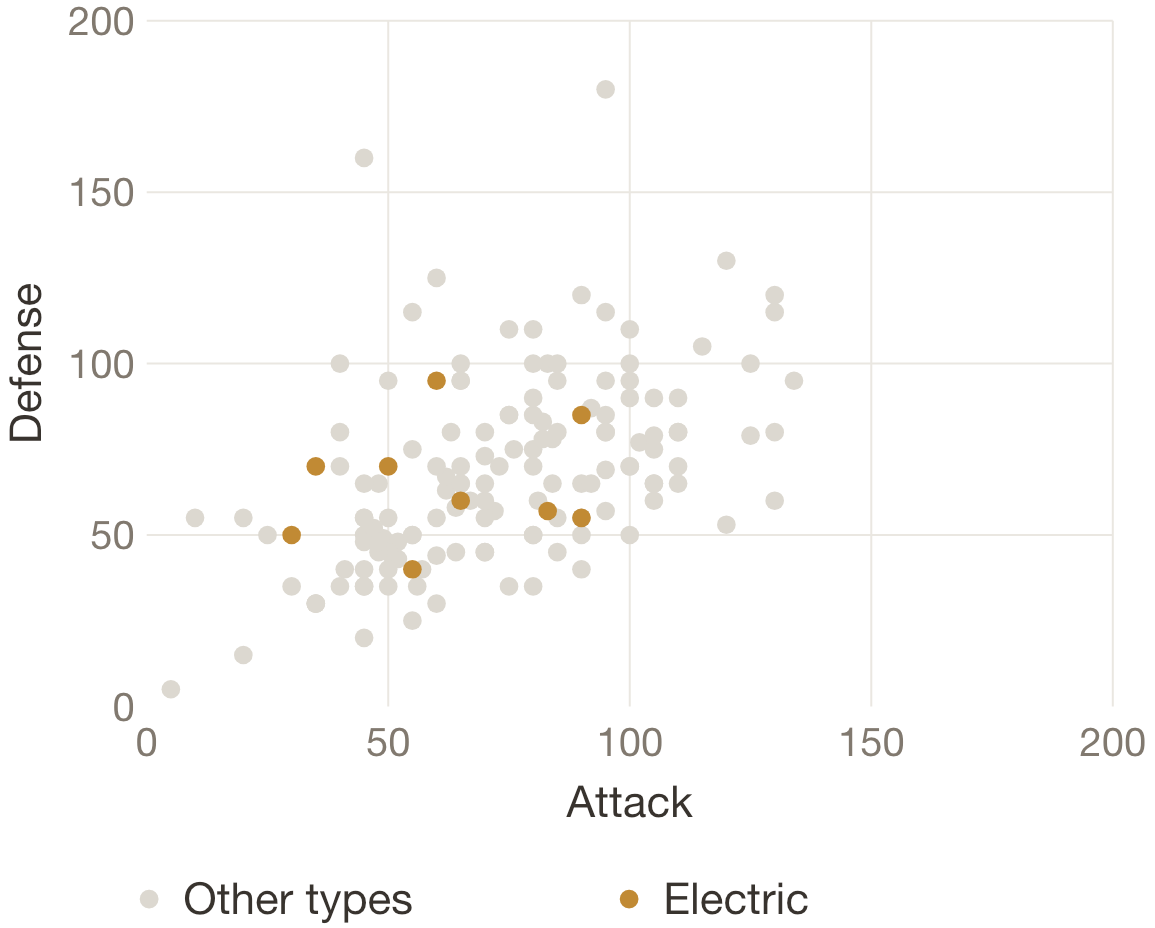
What catches your eye first?

Color everything — or direct attention?

ALL TYPES



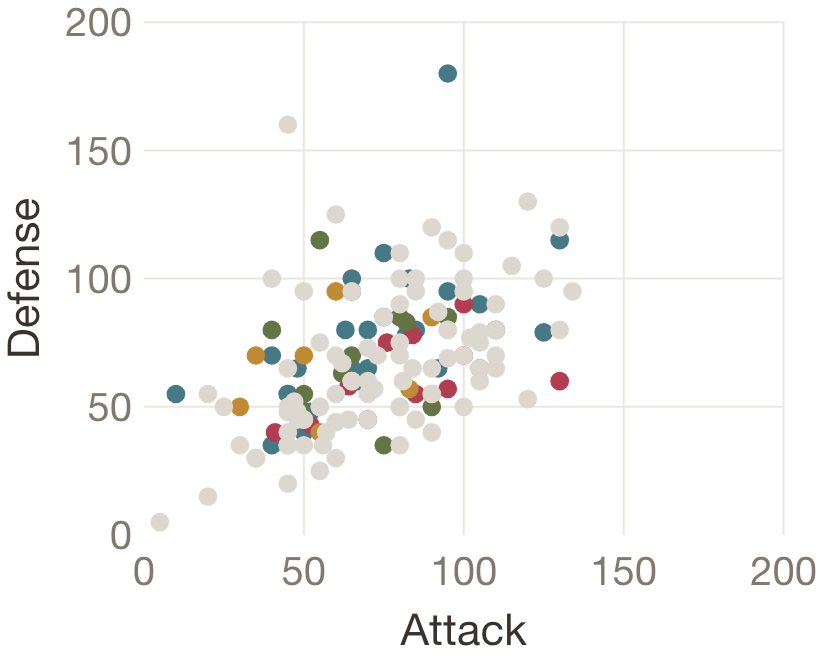
ELECTRIC IN FOCUS



Color needs a job.

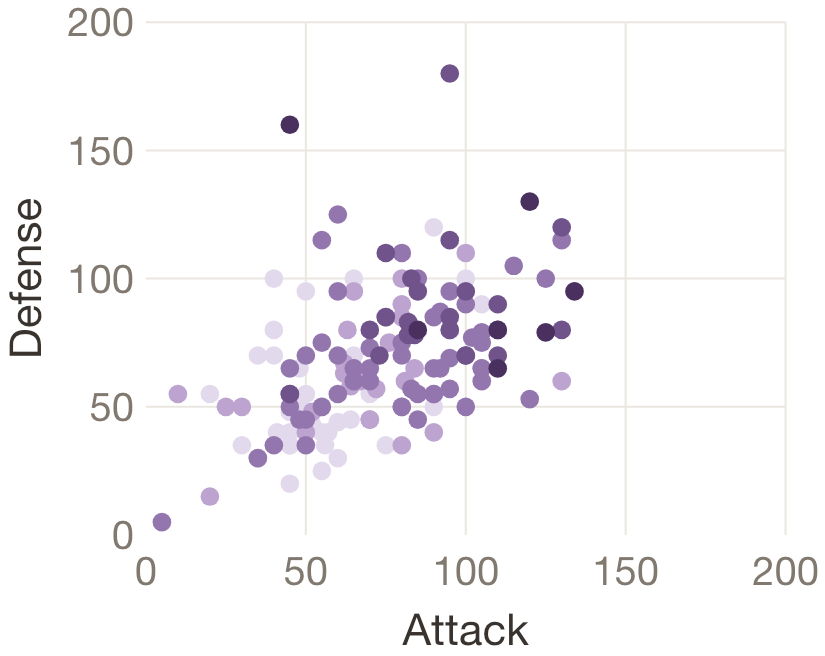
Different kinds of data need different color logic.

TYPE



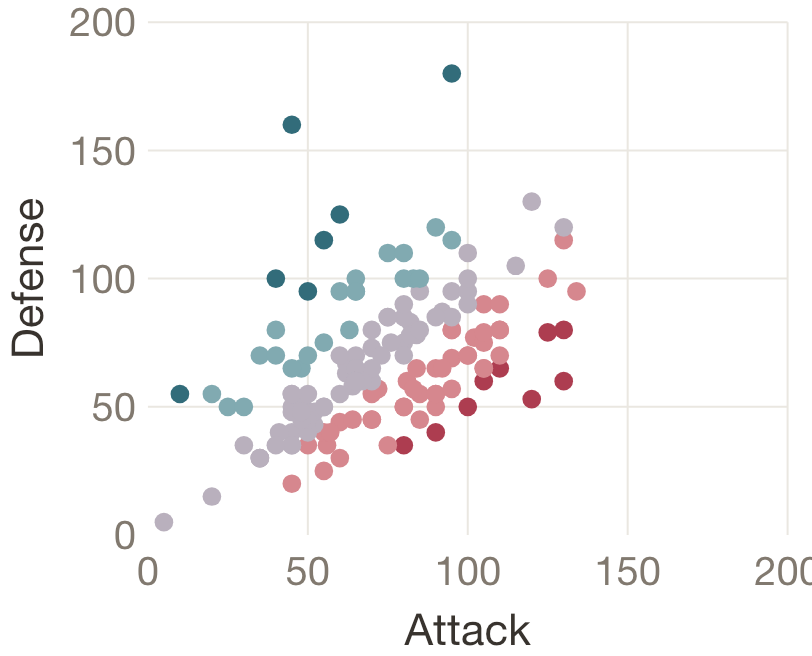
- Water
- Fire
- Grass
- Electric
- Other

WEIGHT



- 0–<10 kg
- 10–<30 kg
- 30–<70 kg
- 70–<150 kg
- 150+ kg

DEFENSE – ATTACK



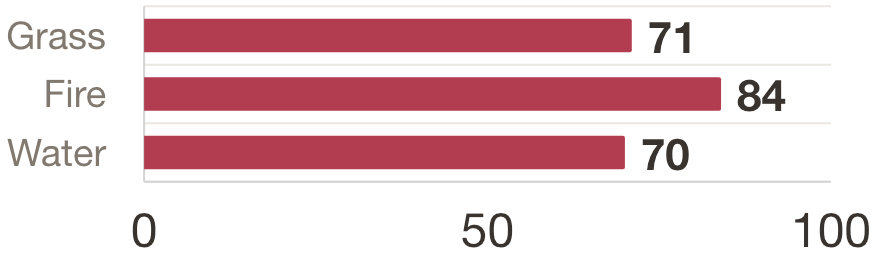
- ≤–41
- –40…–11
- –10…10
- 11…40
- ≥41

Layout creates relationships.

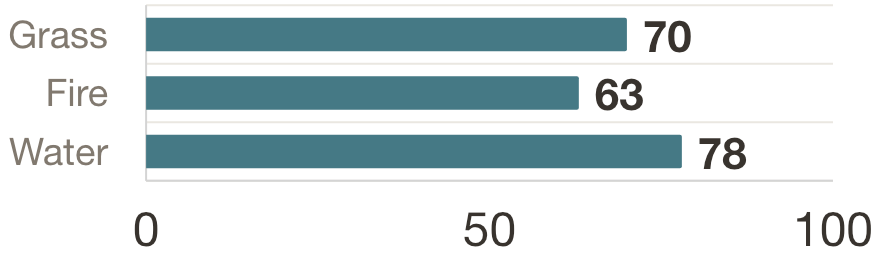
Same values. Different alignment.

SCATTERED

Attack

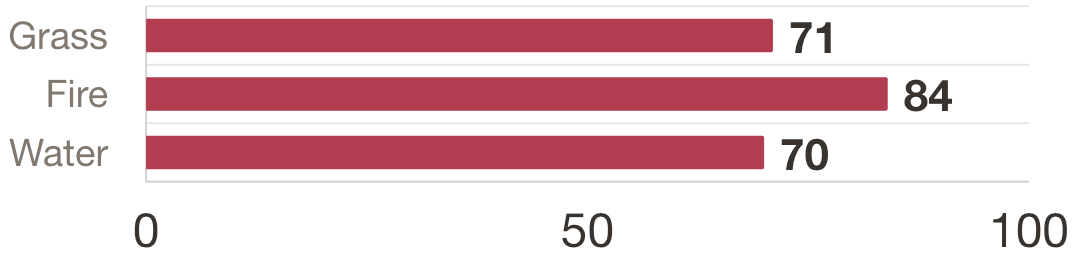


Defense

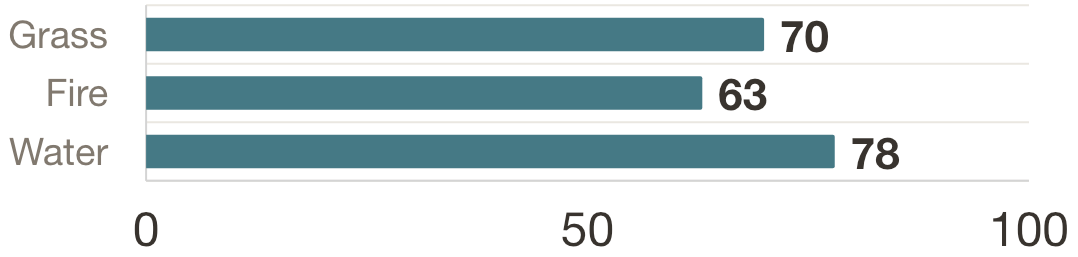


ALIGNED

Attack



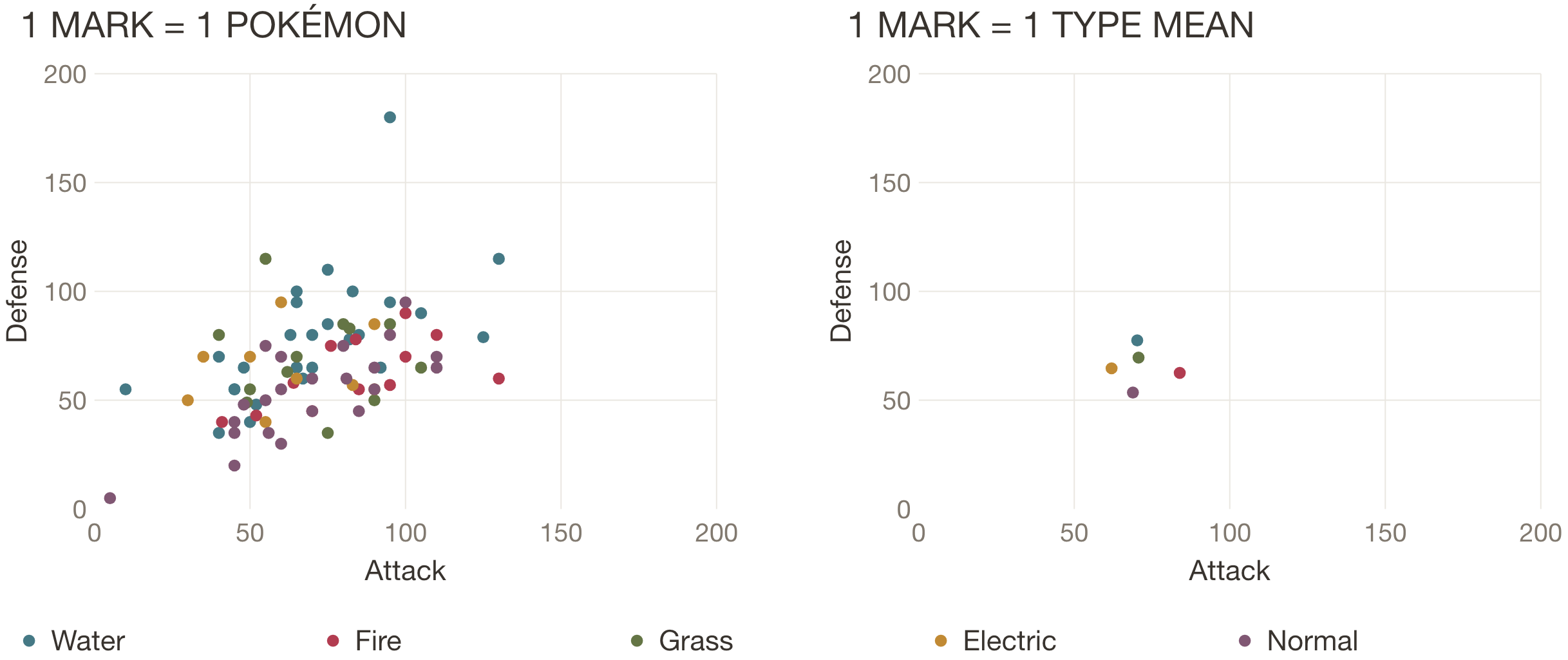
Defense



Mean base stats · primary type · rounded to whole numbers

What does one mark represent?

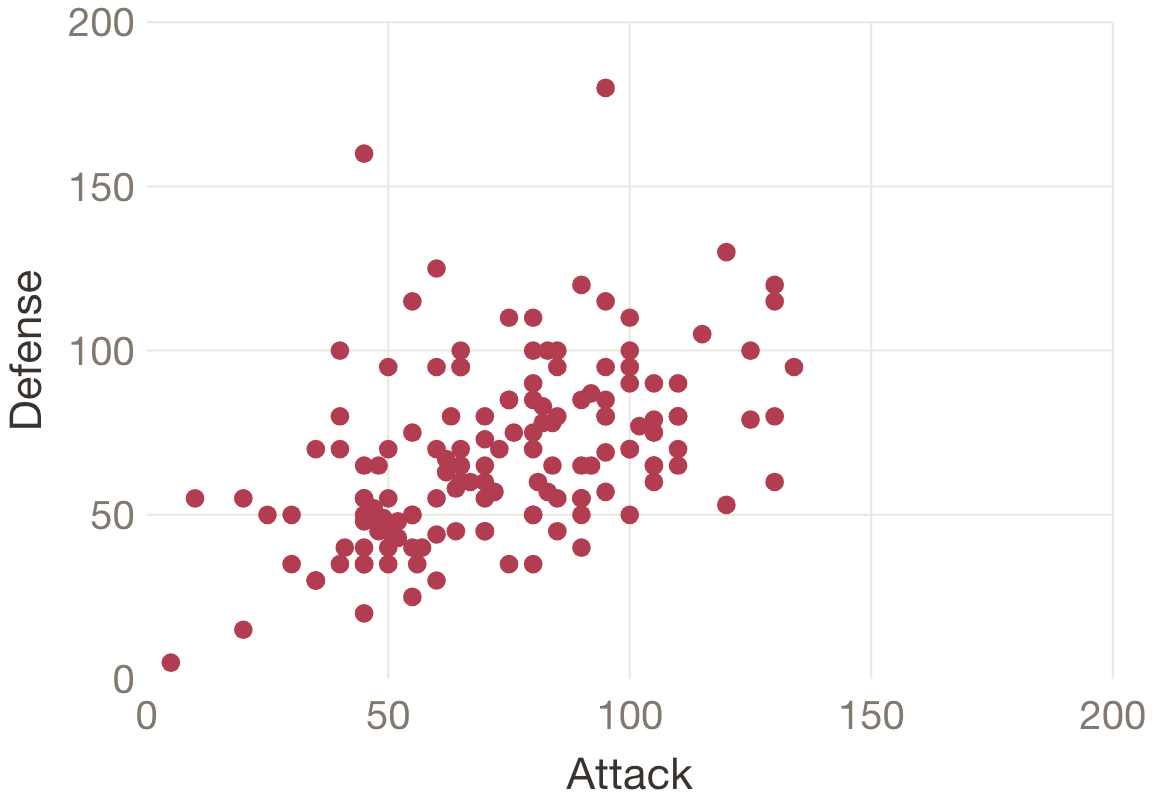
Aggregation changes the story.



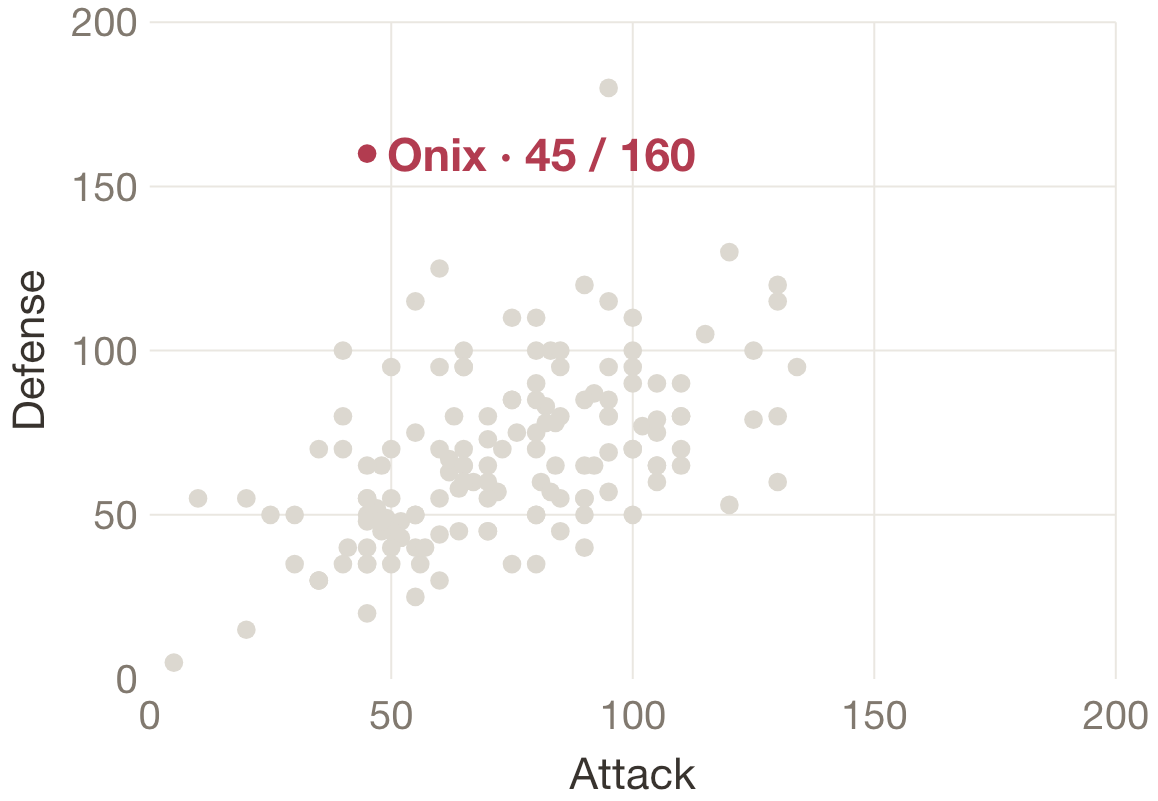
Help the reader find the point.

A title is a chance to say something.

Pokémon stats



Onix: high Defense, low Attack



ONE CLEAR CLAIM. ONE USEFUL LABEL.